

WBS Element	Resources	Weighted Avg Rate, \$/hour	Labor		Type	Materials, Bulk		Tools	Services	Reserves	Budget	Actual	Variance	Comments
			Effort, man- hours	Cost		Cost Per Unit	Cost							
1.1 Project Initiation & Research		44 Hours												
1.1.1 Time Spent for business plan definition		8 Hours												
Identify business problem & Opportunity	PM, Program Mgr	\$136.00	1	\$136.00							\$136.00			
Define strategic alignment	Program Mgr, Exec Sponsor	\$140.00	1.5	\$210.00							\$210.00			
List high-level objectives and success criteria	PM, Stakeholders	\$151.00	1	\$151.00							\$151.00			
Identify constraints and assumptions	PM, Business Analyst	\$100.50	1.5	\$150.75							\$150.75			
Draft & format business case doc	PM, Business Analyst	\$95.50	1.5	\$128.25							\$128.25			
Review & refine	PM, Program Mgr	\$136.00	1	\$136.00							\$136.00			
Send for Sponsor review	PM	\$61.00	0.5	\$30.50							\$30.50			
1.1.2 Develop Project Charter		36 Hours												
Identify Project Objectives	PM	\$61.00	4	\$244.00							\$244.00			
Define high-level scope	PM + Stakeholders	\$75.50	6	\$453.00							\$453.00			
Define major deliverables	PM + SME	\$73.00	5	\$365.00							\$365.00			
Establish high-level timeline	PM	\$61.00	4	\$244.00							\$244.00			
Define success criteria & assumptions	PM + Sponsor	\$75.50	5	\$377.50							\$377.50			
Define success criteria/KPIs	PM + Sponsor	\$95.50	4	\$382.00							\$382.00			
Draft Project charter Document	PM	\$61.00	4	\$244.00							\$244.00			
Internal review	PM + Sponsor	\$75.50	4	\$302.00							\$302.00			
1.2 Planning Phase		88 Hours												
Conduct Stakeholder Interviews	PM, Program Mgr, Stakeholders	\$75.33	48	\$3,616.00							\$3,616.00			
Gather & Document Requirements	PM, Business Analyst	\$57.40	40	\$2,296.00							\$2,296.00			
1.2.3 Finalize requirements & Scope		32 Hours												
Develop final Scope Statement	PM, Business Analyst, Sponsor, Stakeholders	\$73.00	5	\$365.00							\$365.00			
Finalize deliverables	PM, Functional Leads, Devs, Designers, QA, Stakeholders	\$72.00	6	\$432.00							\$432.00			
Finalize High level budget	PM, Finance Rep, Functional Lead, Procurement, PMO	\$76.00	4	\$304.00							\$304.00			
Develop HR Plan	PM, HR Rep, Department Managers, Functional Leads, PMO	\$70.00	3	\$210.00							\$210.00			
Develop Quality Management Plan	QA Leader, PM, Dev Lead, QA Engineers, Business Analyst, Stakeholders	\$67.00	4	\$268.00							\$268.00			
Develop Risk Management Plan	PM, Business Analyst, Tech Leads, QA, Stakeholders	\$68.00	4	\$272.00							\$272.00			
Develop Procurement Management Plan	PM, PMO, Stakeholders, QA Lead, Dev Lead	\$72.00	2.5	\$180.00							\$180.00			
Develop Change Procedures	PM, Functional Leads, SMEs	\$71.00	2	\$142.00							\$142.00			
Develop Schedule	PM	\$61.00	1.5	\$91.50	Jira License			\$210.00			\$91.50			
1.3 Design Phase		120 Hours												
1.3.1 Design System Architecture		80 Hours												
Review Business & Functional requirements	Systems Architect, PM, Tech Leads	\$75.00	12	\$900.00	Adobe Suite			\$3,000.00		\$450.00	\$1,350.00			
Define high-level architecture(MVC, Microservices)	Systems Architect	\$85.00	12	\$1,020.00						\$510.00	\$1,530.00			
Select tech stack (frameworks, tools, DB's, APIs)	Architect, Dev Leads	\$77.50	14	\$1,085.00						\$542.50	\$1,627.50			
Establish system interfaces/integration points	Architect, Integration SME	\$82.50	14	\$1,155.00						\$577.50	\$1,732.50			
Draft architecture diagram and documentation	Architect, Tech Writer	\$73.00	12	\$876.00						\$438.00	\$1,314.00			
Review/approve architecture with stakeholders	PM, Architect, Sponsor	\$92.00	16	\$1,472.00						\$736.00	\$2,208.00			
1.3.2 Create UI Mockups		40 Hours												
Review user requirements and workflows	UX Designer, Business Analyst	\$60.00	6	\$360.00							\$360.00			
Sketch wireframes (low fidelity)	UX Designer	\$65.00	8	\$520.00							\$520.00			
Develop high-fidelity mockups using tools (Figma, Adobe XD)	UX/UI Designer	\$65.00	10	\$650.00	Figma & Adobe			\$6,000.00			\$650.00			
Run usability review / feedback session	UX Designer, PM, Stakeholders	\$72.00	8	\$576.00							\$576.00			
Revise and finalize UI designs	UX Designer	\$65.00	8	\$520.00							\$520.00			
1.4 Development Phase		280 Hours												
1.4.1 Backend Development		160 Hours												
Set up project repo & backend framework (Express, Django)	Dev Lead, Backend Devs	\$70.00	25	\$1,750.00	Github Team Plan			\$4.00		\$875.00	\$2,625.00			
Design & create database schema	Backend Devs, DB Admin	\$70.00	20	\$1,400.00						\$700.00	\$2,100.00			
Develop core API endpoints	Backend Devs	\$70.00	30	\$2,100.00						\$1,050.00	\$3,150.00			
Implement business logic & rules	Backend Devs	\$70.00	25	\$1,750.00						\$875.00	\$2,625.00			
Conduct unit tests & backend validation	Backend Devs, QA Engineer	\$57.50	30	\$1,725.00						\$862.50	\$2,587.50			
Peer review / code review	Dev Lead, Backend Devs	\$70.00	30	\$2,100.00						\$1,050.00	\$3,150.00			
1.4.2 Frontend Development		120 Hours												
Set up frontend repo & environment (React, Angular, etc)	Dev Lead, Frontend Devs	\$70.00	15	\$1,050.00						\$525.00	\$1,575.00			
Integrate approved UI mockups	Frontend Devs	\$70.00	20	\$1,400.00						\$700.00	\$2,100.00			
Develop component-based UI (forms, dashboards)	Frontend Devs	\$70.00	25	\$1,750.00						\$875.00	\$2,625.00			
Connect frontend APIs	Frontend Devs	\$70.00	20	\$1,400.00						\$700.00	\$2,100.00			
Conduct unit & integration testing	Frontend Devs, QA Engineer	\$57.50	20	\$1,150.00						\$575.00	\$1,725.00			
Peer review / code review	Dev Lead	\$70.00	20	\$1,400.00						\$700.00	\$2,100.00			

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1.5 Testing Phase		160 Hours												
1.5.1 Unit Testing		40 Hours												
Write unit tests for backend functions	Backend Devs	\$70.00	10	\$700.00						\$350.00	\$1,050.00			
Write unit tests for frontend components	Frontend Devs	\$70.00	10	\$700.00						\$350.00	\$1,050.00			
Run test suites & verify coverage	Devs, QA Engineer	\$57.50	10	\$575.00						\$287.50	\$862.50			
Review failed tests & fix small logic bugs	Devs	\$70.00	10	\$700.00						\$350.00	\$1,050.00			
1.5.2 User Acceptance Testing		80 Hours												
Develop UAT scripts & scenarios	QA Lead, PM, Stakeholder	\$75.00	20	\$1,500.00						\$750.00	\$2,250.00			
Schedule UAT sessions with end users	PM, Functional SME	\$73.00	10	\$730.00						\$365.00	\$1,095.00			
Facilitate walkthroughs & collect feedback	QA, PM, Stakeholders	\$72.00	25	\$1,800.00						\$900.00	\$2,700.00			
Track & document issues	QA Lead	\$45.00	25	\$1,125.00						\$562.50	\$1,687.50			
1.5.3 Fix Bugs from UAT		40 Hours												
Triage reported bugs & prioritize	QA Lead, Dev Leads	\$57.50	10	\$575.00						\$287.50	\$862.50			
Fix issues & re-test	Backend & Frontend Devs	\$70.00	15	\$1,050.00						\$525.00	\$1,575.00			
Conduct final regression testing	QA	\$45.00	10	\$450.00						\$225.00	\$675.00			
Prepare go-live checklist update	PM, QA	\$53.00	5	\$265.00						\$132.50	\$397.50			
1.6 Deployment		72 Hours												
1.6.1 Prepare Deployment Plan		24 Hours												
Draft detailed deployment checklist	PM, Dev Leads, QA Lead	\$68.00	6	\$408.00						\$204.00	\$612.00			
Define rollback/contingency plan	DevOps, QA	\$57.50	6	\$345.00						\$172.50	\$517.50			
Validate deployment approach with stakeholders	PM, Tech Leads, Sponsor	\$75.00	6	\$450.00						\$225.00	\$675.00			
Assign responsibilities for deployment activities	PM, Functional Managers	\$61.00	6	\$366.00						\$183.00	\$549.00			
1.6.2 Migrate Data		16 Hours												
Perform dry-run data migration	DB Admin, DevOp, QA	\$77.50	4	\$310.00						\$155.00	\$465.00			
Validate migrated data integrity	QA, SMEs	\$65.00	4	\$260.00						\$130.00	\$390.00			
Finalize live migration schedule	PM, DevOps	\$66.00	4	\$264.00						\$132.00	\$396.00			
Execute production data migration	DevOps, DB Admins	\$77.50	4	\$310.00						\$155.00	\$465.00			
1.6.3 Go/No-Go Meeting		8 Hours												
Review UAT sign-off and risk register	PM, QA Lead	\$53.00	2	\$106.00							\$106.00			
Present deployment readiness checklist	PM, Dev Leads, QA	\$60.00	2	\$120.00							\$120.00			
Document approvals and escalation plan	PM, Sponsor	\$96.00	2	\$192.00							\$192.00			
Record official Go/No-Go decision	PM, Stakeholders	\$75.00	2	\$150.00							\$150.00			
1.6.4 Go Live		24 Hours												
Deploy production codebase	DevOps, Engineers	\$70.00	6	\$420.00							\$420.00			
Monitor system for critical errors	DevOps, QA	\$57.50	6	\$345.00							\$345.00			
Provide hypercare & real-time support	QA, Support Team	\$55.00	6	\$330.00							\$330.00			
Confirm production stability checklist	PM, QA, Tech Leads	\$65.00	6	\$390.00							\$390.00			
1.7 Close-Out		56 Hours												
1.7.1 Post-Launch Support		40 Hours												
Monitor for defects or production issues	QA, Support Analysts, DevOps	\$57.50	10	\$575.00							\$575.00			
Address post-launch tickets	Developers, Support Staff	\$57.50	10	\$575.00							\$575.00			
Conduct performance tuning if needed	Dev Leads, DB Admin	\$77.50	10	\$775.00							\$775.00			
Transition ownership to operations or help desk team	PM, Support Manager, Sponsor	\$89.00	10	\$890.00							\$890.00			
1.7.2 Collect Lessons Learned		16 Hours												
Schedule and conduct lessons learned meeting(s)	PM, Key Stakeholders	\$75.00	4	\$300.00							\$300.00			
Gather feedback from team and stakeholders	PM, All Contributors	\$65.00	4	\$260.00							\$260.00			
Document wins, blockers, and proposed improvements	PM, Business Analyst	\$58.00	4	\$232.00							\$232.00			
Store documents in project repository	PMO Admin, PM	\$50.00	4	\$200.00							\$200.00			
		TOTALS	852	\$58,582.50						\$19,183.00	\$77,765.50			